Information sheet

Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race

Regular research paper

1. What is the main claim, and why does it matter to the autonomous agents and multi-agent systems literature?

In an ecosystem of deployed agents, the payoff an agent earns and the quantity that decides whether its design is replicated are different functionals. The agent plays; a principal who is not a player decides whether to keep deploying it, and internalises only a fraction λ of the harm the design causes third parties. We call that gap the *selection wedge*. Writing a for the expected task payoff of an interaction, m for its expected number of unsafe actions and h for the harm each one causes, the population of designs evolves under the replicator equation of the principal's functional $\pi_P = a - \lambda h\, m$ rather than of the social one. That functional depends on pass-through and harm only through the single product $L = \lambda h$, and it makes every invasion condition affine in L. The claim is that governance of such an ecosystem binds at the layer that selects designs and not at the layer that plays them: an instrument applied to the agents can be exactly inert, while the instrument that does move the vector field is not monotone in its own strength.

This matters to the autonomous agents and multi-agent systems literature because that literature builds, evaluates and deploys populations of agents that principals choose, and every instrument it has for open agent populations – social laws, electronic institutions, norm synthesis and sanctioning, trust and reputation, reward transfers and formal contracts – acts from inside the interaction and so moves the behaviour of a population whose membership is given. Its evolutionary models of AI races, institutional incentives and delegation identify earned payoff with reproductive fitness, using one payoff matrix twice. Deployed ecosystems break that identification structurally, and the two-matrix version yields guidance the one-matrix version cannot express: which admission tests are dynamically empty and what to replace them with, why a population that looks stationary can be losing its protection, and why an accountability instrument has a range of strengths worse than a weaker setting.

2. What is the evidence, and what does it imply?

The interaction layer is a four-design race – always safe AS, always unsafe AU, conditionally safe CS, conditionally antisocial CAS – taken unchanged from the framed AI-race experiment of Fernández Domingos and Han (preprint, arXiv:2607.26034, July 2026) and evaluated by exact expectation over the horizon law rather than by sampling. An invader beats a resident when its payoff and unsafe-action advantages satisfy $\delta a - L\,\delta m > 0$, so each of the twelve ordered invasions has a critical liability $L^* = \delta a/\delta m$ wherever the two designs differ in unsafe count. Four of these, together with one interior threshold, organise the paper and span a factor of 368: $0.116 < 0.551 < 4.274 < 42.765$. Each is a property of π_P alone and is therefore shared by every imitative payoff-monotone dynamics, which we confirm for the logit, best-response and aspiration rules.

Solo evaluation is dynamically empty. In this pool CAS weakly dominates AU at every $L \geq 0$, so a filter that scores a design against a safe environment removes only a weakly dominated design; the attractor it deletes carries the maximal long-run unsafe frequency $U = 1$, and below $L = 0.116$ the surviving pool reproduces that same all-unsafe state on a different design. Filtered and unfiltered long-run unsafe frequencies agree to within 1.2×10^{-3} on a common start measure. The dominance is a property of the pool, not a theorem about filters: it fails once the pool contains a design that punishes for longer than it is punished, which is the one design-layer intervention the selection layer does not undo. Over all fifteen probe sets the separating margin takes exactly two values, 0.539 when the probe reciprocates and contains no always-unsafe design and 0.132 otherwise, so a reciprocating probe needs 7 episodes at zero execution error and 16 at a 5% rate where a solo probe needs 111 and 136. The implication is that an evaluation protocol should be specified by who the agent is evaluated against, not by how strict the threshold is.

Protection is governed by a variable that drifts. AS and CS are exact payoff twins, so the safe face carries no selection gradient, yet the liability needed to repel an invader falls from 42.765 at the unconditional vertex to 0.551 at the conditional one, a factor of 77.7 at the baseline calibration and between 4.4 and 90.4 across the robustness grid. Nothing observable in the interaction changes as the population drifts along that face, so resilience has to be monitored as a composition rather than as an outcome.

Long-run harm is not monotone in liability. A mixture-free invasion lemma, valid for any symmetric matrix game, gives the guard threshold $L^\dagger = 4.274$ exactly. On $(4.274, 42.765)$ liability taxes the retaliation that conditional safety depends on, the unsafe rest point becomes uninvadable by conditional safety, and the basin-averaged unsafe frequency jumps discontinuously from 0 to 0.32 as L crosses the lower edge from below. In a finite population the resulting trap is long-lived, with mean escape times from 65 generations at $Z = 30, \beta = 0.02$ to 3.9×10^{13} at $Z = 100, \beta = 0.2$. A dangerous range of this kind exists at all 24 prize and risk grid points, under all five charge schedules we tried, and in every enlarged design pool at zero execution noise; its location is not transferable, since the lower edge moves from 4.274 in the reduced pool to between 12 and 13 in the full one, and by an execution error rate of 5% the range has closed because the state liability was protecting has itself stopped being safe. Assortative matching shrinks the range from above and closes it at $r = 0.354$, whereas misattributing a fraction q of the blame pushes the upper edge up exactly as $1/(1-q)$ while moving the lower edge by at most 11.8%, so at $q = 0.5$ the dangerous range has doubled. The implication for a regulator is that raising liability out of a protected regime has to clear the whole range in one step, and that incident records which book an interaction against the wrong party widen the range they have to clear.

Losing a protected regime costs more than keeping it. Coupling enforcement to a monotone environmental state that erodes it to a residual fraction ρ makes the transition hysteretic: the ratio of the liability at which the regime is recovered to the one at which it was lost is exactly $1/\rho$, for every erosion channel and every time-scale separation, provided the protected state is exactly safe. The cheap intervention is the early one.

3. Closest related contributions by other authors

Evolutionary models of AI races are the closest work: Han, Pereira, Santos and Lenaerts, *Journal of Artificial Intelligence Research* **69** (2020) 881–921; Cimpeanu et al., *Scientific Reports* **12** (2022) 1723; Alalawi et al., *Applied Mathematics and Computation* **508** (2026) 129627; surveyed in Han, *AI Communications* **35** (2022) 327–337. There the racing actor and the reproducing actor coincide. We keep their race and replace the selection layer, so the contribution is orthogonal rather than corrective.

Evolutionary methods and normative design in multi-agent systems. Evolutionary dynamics are a standard tool of this journal's literature: Bloembergen, Tuyls, Hennes and Kaisers, *Journal of Artificial Intelligence Research* **53** (2015) 659–697; Tuyls et al., *Autonomous Agents and Multi-Agent Systems* **34** (2020) 7; Phelps, McBurney and Parsons, *Autonomous Agents and Multi-Agent Systems* **21** (2010) 237–264. Norm and institution design supplies the instruments we compare against: Shoham and Tennenholtz, *Artificial Intelligence* **73** (1995) 231–252; Morales, Wooldridge, Rodríguez-Aguilar and López-Sánchez, *Autonomous Agents and Multi-Agent Systems* **32** (2018) 635–671; Morris-Martin, De Vos and Padget, *Autonomous Agents and Multi-Agent Systems* **33** (2019) 706–749. All of these act on agents inside the interaction; our instrument acts on a principal outside it and changes which designs exist.

Evaluation and reward modification in multi-agent settings. Our evaluation result speaks to Omidshafiei et al., *Scientific Reports* **9** (2019) 9937, which ranks agents by an evolutionary process over the interaction rather than by a solo score, and our non-monotone instrument contrasts with Willis, Du, Leibo and Luck, *Autonomous Agents and Multi-Agent Systems* **38** (2024) 49, and Haupt, Christoffersen, Damani and Hadfield-Menell, *Autonomous Agents and Multi-Agent Systems* **38** (2024) 51, whose transfers and contracts act among the agents themselves and admit a computable minimum. The closest architecture is Zheng, Trott, Srinivasa, Parkes and Socher, *Science Advances* **8** (2022) eabk2607, whose outer planner optimises an instrument over a responding population; ours only selects.

The indirect evolutionary approach: Alger and Weibull, *Econometrica* **81** (2013) 2269–2302, surveyed in *Annual Review of Economics* **11** (2019) 329–354; Dekel, Ely and Yilankaya, *The Review of Economic Studies* **74** (2007) 685–704; Heifetz, Shannon and Spiegel, *Journal of Economic Theory* **133** (2007) 31–57. Ours is the mirror image: their distortion is endogenous and sits in behaviour, ours is an exogenous policy variable inside the selection functional.

Multilevel selection: Wilson, *Proceedings of the National Academy of Sciences* **72** (1975) 143–146; Traulsen and Nowak, *Proceedings of the National Academy of Sciences* **103** (2006) 10952–10955. Ours is degenerate but sharper: one selecting level, not the level that plays, selecting on a policy variable rather than on an aggregate of the lower level.

Multitask incentive distortion: Holmström, *The Bell Journal of Economics* **10** (1979) 74–91; Holmström and Milgrom, *Journal of Law, Economics, and Organization* **7** (1991) 24–52; Baker, *Journal of Political Economy* **100** (1992) 598–614. Liability is exactly such a distorted measure, taxing retaliation along with initiation, but here the reallocation is evolutionary rather than deliberate, and the loss is discontinuous in the power of the instrument rather than smooth.

Game-environment feedback: Weitz et al., *Proceedings of the National Academy of Sciences* **113** (2016) E7518–E7525; Tilman, Plotkin and Akçay, *Nature Communications* **11** (2020) 915. Their environmental state can recover and deployed capability cannot, which is what makes our hysteresis width exactly computable. Nearby, Fernández Domingos et al., *Scientific Reports* **12** (2022) 8492, and Terrucha et al., *Scientific Reports* **14** (2024) 10460, treat delegation as a commitment device chosen inside the game; our principal sits outside it.

4. Has any part been published before?

No. No part of this paper has been published or presented before, at a conference or elsewhere, and it is under review nowhere else. There is no earlier version by the authors as a workshop paper, extended abstract or preprint.

The one component we did not originate is the interaction layer, which is taken unchanged from the framed AI-race experiment of Fernández Domingos and Han (preprint, arXiv:2607.26034, July 2026), by different authors: its stage game, uncertain horizon, four reduced designs and parameter values are reused as that preprint reports them. The reuse is deliberate, so that every result here is attributable to the deployment layer alone; because every threshold is a closed form in the payoff matrix, the results carry over unchanged in form to any revised version of those constants. Computations use EGTtools (Fernández Domingos, Santos and Lenaerts, *iScience* **26** (2023) 106419). All code, tables and figures are public under the MIT licence at `https://github.com/trungkiet2005/deployment-layer-selection`, and every number quoted in the manuscript is regenerated by the released scripts and pinned by its test suite.